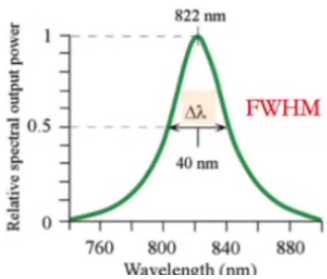
Marmara University
Electrical and Electronics Engineering Department
Spring 2023
EE 4091 - Photonics
Assignment (3) - Due June 9, 2023

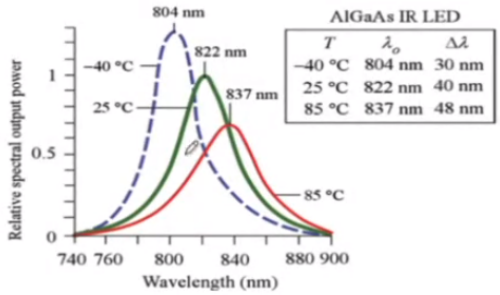
Part I: LEDs

1) The emission spectrum for an AlGaAs IR LED is given in the figure for three different temperatures. Verify the peak wavelength and spectral linewidth vs. *T* relations for this case.
2) It is found that approximately 2.5 mW light output power is collected from a particular LED when the current is 50 mA and the voltage is 1.6 V. Calculate the LED external efficiency.

1) Spectral Distribution of an LED is described by peak wavelength λ_o and spectral width $\Delta\lambda$.

2 kağıtta





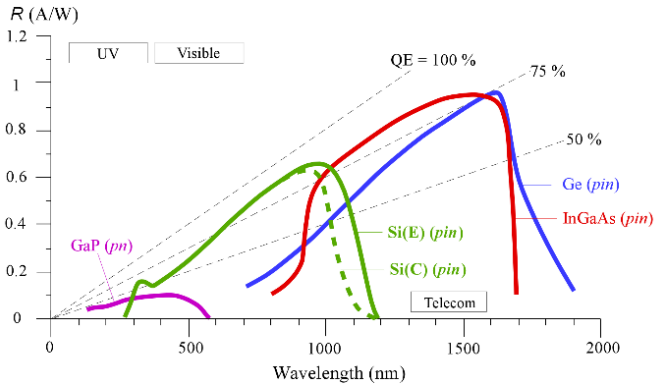
Part II: Lasers

1) The cavity of a He-Ne laser is 1 m long with mirror reflectivities of 0.99. The laser operates at 632.8 nm. What is the mode number and the free spectral width of this laser? Also, find the quality factor and finesse of the cavity. Assume the index of refraction is n=1.

2) Design the cavity of an InGaAsP (n=3.45) for a single mode laser emitting around 1300 nm wavelength. The laser gain spectrum has a width of 8.8 nm. Comment on your result

Part III: Photo-diodes

Consider the commercial InGaAs, Ge, Si (Enhanced and Conventional) pin photodiodes. The responsivity curves are given in the figure below.



1) For each photodiode, find the optical power at 1550 nm that will give a photocurrent of 10 nA. Comment on the answers.
2) What would be the QE for each photodiode in part (1)? Comment on the answers.
3) For each photodiode, for the same optical power in part (1), find the photocurrent at 1064 nm.
4) Plot the QE of the InGaAs pin photodiode versus wavelength. Comment on the figure.

Show all work and analysis
Only pdf files are acceptable
Late submissions are checked but not graded
Copied submissions represent academic dishonesty and are disqualified